

FieldSense FT-450 Magnetic Flow Transmitter -- Technical Manual

PRODUCT OVERVIEW

The FieldSense FT-450 is a state-of-the-art magnetic flow transmitter designed for precise monitoring of flow rates in water and wastewater treatment applications. Its robust design and advanced technology make it ideal for measuring plant inlet/outlet flow, chlorine and coagulant dosing flow, and treated effluent flow. The FT-450 employs electromagnetic principles to deliver accurate flow measurements, ensuring optimal process control and efficiency in treatment plants.

Key features of the FT-450 include a user-friendly interface, versatile installation options, and compatibility with various process media. Whether used in a municipal water treatment facility or an industrial wastewater plant, the FT-450 is engineered for reliability and durability in demanding environments.

TECHNICAL SPECIFICATIONS

- Flow Range: 0.1 to 10,000 m³/h
- Accuracy: ±0.5% of reading
- Electrode Material: Hastelloy C-276 or 316 Stainless Steel
- Process Connection: Flanged (ANSI 150), welded, or threaded options available
- Output Signal: 4-20 mA, pulse output, or Modbus RTU
- Ingress Protection: IP68 rated for submerged applications, IP65 for general use

CALIBRATION PROCEDURE

Calibration of the FieldSense FT-450 is essential to maintain measurement accuracy. The calibration process involves two primary methods: empty-pipe calibration and full-scale calibration.

****Empty-Pipe Calibration:****

1. Ensure the FT-450 is powered off.
2. Remove the flow transmitter from the pipeline.
3. Connect the FT-450 to the calibration device as per the manufacturer's instructions.
4. Fill the calibration device with water and ensure there is no air in the system.
5. Power on the FT-450 and set it to calibration mode via the configuration menu.
6. Run the calibration device at various flow rates (0.1 m³/h to 10 m³/h), recording the FT-450 output for each setting.
7. Once complete, confirm the calibration data and save the settings.

****Full-Scale Calibration:****

1. Install the FT-450 back on the pipeline, ensuring all connections are secure.
2. Adjust the flow rate in the pipeline to the desired maximum flow.
3. Connect a reliable flow measurement device in parallel to the FT-450.
4. Allow the flow to stabilize, then compare the readings from both devices.
5. If discrepancies are noted, adjust the calibration settings in the FT-450 until the readings align within the

acceptable accuracy range.

6. Save the calibration settings and exit calibration mode.

ERROR CODES

The FT-450 features a built-in diagnostic system that alerts users to potential issues. Below are common error codes, their causes, and recommended actions:

- ****E001: No Flow Detected****

Cause: The flow rate is below the minimum threshold.

Resolution: Check for blockages or ensure the flow is above 0.1 m³/h.

- ****E002: Electrode Fault****

Cause: Faulty or damaged electrodes.

Resolution: Inspect electrodes for damage, clean as necessary, or replace if damaged.

- ****E003: Calibration Error****

Cause: Calibration settings are incorrect or not saved.

Resolution: Recheck calibration steps and ensure settings are saved properly.

- ****E004: Signal Loss****

Cause: Loss of signal from the flow measurement.

Resolution: Check connections and wiring for integrity; ensure no loose connections.

- ****E005: Power Supply Error****

Cause: Insufficient voltage supplied to the unit.

Resolution: Verify the power supply meets the required specifications (24V DC).

MAINTENANCE

Regular maintenance of the FT-450 is crucial for optimal performance and longevity. Key maintenance procedures include electrode cleaning and troubleshooting.

****Electrode Cleaning:****

1. Power off the FT-450 and isolate the unit from the process.
2. Remove the transmitter from the pipeline.
3. Use a soft cloth and a mild detergent solution to clean the electrodes.
4. For heavy deposits, a soft brush may be used, taking care not to scratch the electrode surfaces.
5. Rinse thoroughly with clean water and allow to dry before reinstallation.

****Troubleshooting:****

- If the FT-450 displays error codes or inaccurate readings, follow the error code guidelines provided.
- Check for any obstructions within the pipeline that may affect flow.
- Verify all connections and wiring are intact and secure.
- If issues persist, consult the FieldSense technical support team for advanced diagnostics.

By adhering to the procedures outlined in this manual, operators can ensure that the FieldSense FT-450 maintains its high performance and accuracy in flow measurement applications.